EXPERIMENT NO. 2

FARADAY EFFECT USING SODIUM LAMP

I Introduction

Michael Faraday discovered in 1845 that a transparent block of glass when viewed in the direction of a strong applied magnetic field rotates the vibration direction of the incident light waves. The glass is said to have become optically active. Since then this phenomenon has been observed in various solids, liquids and gases (Jenkins & White, p 596-600). This effect is analogous to Zeeman effect. The amount of rotation of the vibration direction of the incident light is given by the expression

$$\theta = VBL \quad , \tag{1}$$

where θ is the angle of rotation measured in minutes of arc, V is a constant known as the Verdet constant, B is the magnetic field in Tesla and L is the optical path length through the material measured in 'm'. The aim of the present experiment is to determine Verdet's constant for the two given transparent blocks of glass (light flint and extra dense flint). The rotation is measured directly. The magnetic field is measured by means of a flip coil and a ballistic galvanometer.

Physically this effect arises from the differences in propagation of the two senses of circular polarization into which linearly polarized waves may be resolved. The Faraday rotation per unit length θ is

$$\theta = -\frac{\omega}{2c}(n_+ - n_-) \quad , \tag{2}$$

where c is the speed of light, n_+ and n_- are the indices of refraction of right and left circularly polarized radiation of frequency ω. Classically treating the interaction of an electron with an electromagnetic wave under the influence of a dc magnetic field, we have

$$\theta = \frac{e}{2mc^2}\frac{B}{\mu_o}\lambda\frac{dn}{d\lambda} \qquad \text{and} \qquad n^2 = 1 - \frac{Ne^2}{\varepsilon_o m\omega^2} \quad , \tag{3}$$

where n is the mean index of refraction for linearly polarized light, N is the carrier concentration and m is the mass of the electron. The experiment consists in calibrating the electromagnet and subsequently measuring the Verdet constant for glass blocks.

II Experimental Procedure

A. Calibration of the galvanometer

The circuit is shown in Fig.1. A capacitor of known capacitance is charged from a storage cell SC of known voltage. It is discharged through the galvanometer by means of a two-way key (the so-called SPDT switch). The deflection, d in mm, of the galvanometer spot, the charge Q_g flowing through the galvanometer and the calibration constant K of the galvanometer are related by

$$Q_g = Kd \qquad \text{or,} \qquad K = (Q_g/d) \text{ Coulomb/mm.} \tag{4}$$

As the damping and subsequent deflection of the galvanometer are strongly affected by the external resistance one must remember to calibrate the galvanometer with the same external resistance R_e used with the flip coil and make correction in d due to damping as explained in Expt. No. 5 (B-H curve, section III B). Otherwise the results will be erroneous. The circuit used for calibration is shown in Fig.1. Then

$$Kd = Q_g = CV\,(R_e + R_f)\,/(R_e + R_f + R_g)\ , \qquad (5)$$

where R_g = galvanometer resistance, R_e = external resistance, R_f = flip coil resistance, L = flip coil inductance (not used in Eq. (5)), C = known capacitance, and V = storage cell voltage.

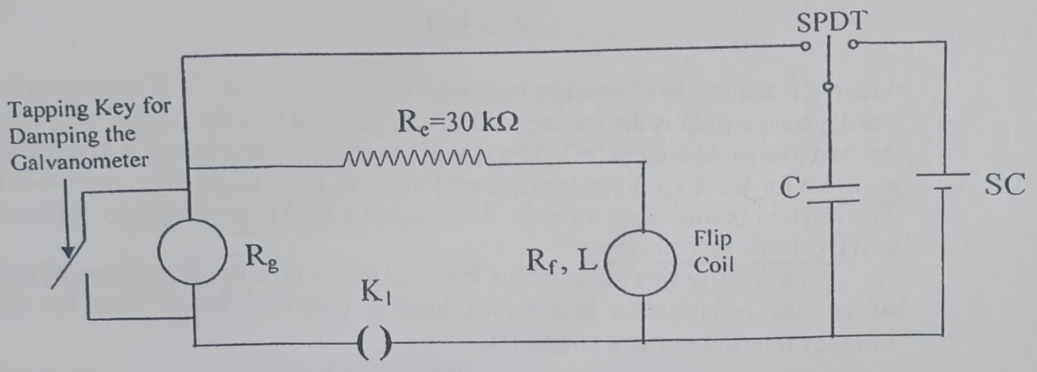


Fig. 1

R_e should be adjusted so that there is a large deflection (almost full scale) when discharging C. If the external resistance in the circuit is kept equal to the critical damping resistance of the galvanometer, the galvanometer spot will return to zero fast enough to allow measurements to be made conveniently (Worsnop & Flint, p 510-514). Use two capacitors of different values and their combinations for this part to find an average value of K.

B. Measurement of the magnetic field using a flip coil

Set up the circuit as shown in Fig.2.

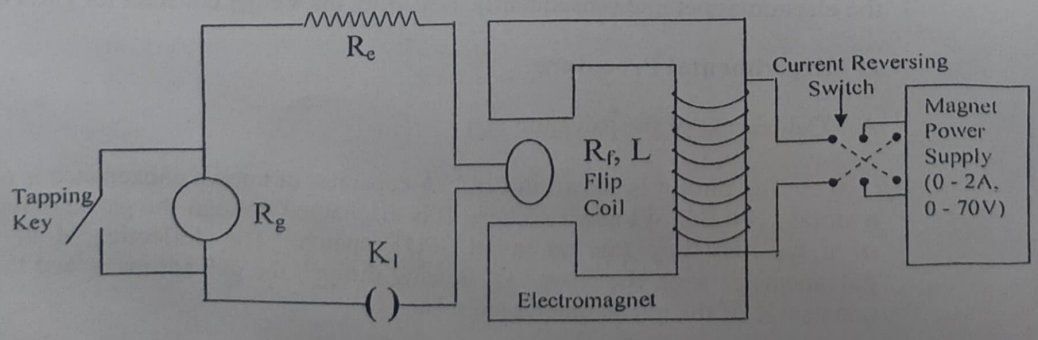


Fig. 2

The flip coil consists of an accurately wound coil of hair-fine wire which is fixed at the end of an insulating rod. It is, therefore, necessary to take the precaution of handling the coil very carefully. Rough handling will damage the layer of resin with which the coil is impregnated. The flip coil is placed between the pole-pieces of an electromagnet whose terminals are connected to a power supply (0-2A, 0-70V) in series with an ammeter and a current reversing switch. A knob allows the coil to be flipped through 180° which causes a charge Q to flow through the galvanometer subsequently producing a deflection d in the galvanometer. We then have

$$\phi = NAB \ , \tag{6}$$

where ϕ = total flux through the coil, N is its number of turns, A is its cross-section in cm^2 and B is the magnetic field in gauss. The charge q flowing through the circuit is given by (note that i = dq/dt)

$$L\frac{di}{dt} + R\frac{dq}{dt} = -10^{-8}\frac{d\phi}{dt} = -10^{-8}NA\frac{dB}{dt} \ . \tag{7}$$

or, $$L\int_0^0 di + R\int_0^Q dq = 10^{-8}NA\int_{-B}^{+B} dB$$ (Sign is immaterial),

or, $$RQ = 2\times 10^{-8}NAB \ ,$$

or, $$B(gauss) = 10^{8}\frac{RK}{2NA}d \ . \tag{8}$$

Using the galvanometer constant K determined earlier, the deflection d, the resistance $R = R_e + R_f + R_g$ (refer to Fig.1) and NA = the product of the number of turns and area of the flip coil, the magnetic field B can be calculated from Eq.(8). Tabulate the results and plot a graph of the magnetic field vs. the current (0-1.5A in steps of 0.1A) through the electromagnet.

The flipping motion of the flip-coil must be smooth but fast enough for accurate measurement of B. A jerky or slow flipping action will lead to errors. Make several trials for flipping and note the corresponding values of d for a given magnet current. Make correction in d due to damping as mentioned in section IIA above. Calculate B and find its average value for the set current. Now reduce the current to zero smoothly & slowly. Switch off the power supply and make use of the four way key (commutator) to change the direction of current into the magnet coils. Switch on the power supply and smoothly increase the current to the same value. Measure B for the reverse current and plot.

III Determination of the Verdet Constant for the given glass blocks (light flint & dense flint)

The sodium lamp source along with a polarizer is aligned to lie along the axis of the partially hollow magnet pole-pieces. Place the glass block securely between the poles of the electromagnet. AVOID ACCIDENTAL FALL OF GLASS BLOCKS. The terminals of the electromagnet are connected to the power supply in series with a DC ammeter. ONE MUST MAKE SURE THAT THE CURRENT CONTROL KNOB OF THE POWER SUPPLY IS

KEPT AT ZERO BEFORE SWITCHING ON/OFF THE POWER. Failing to do so will totally damage the equipment. Now the power is switched on and the power supply is allowed to warm up for a few minutes. The current control knob of the unit is then slowly turned to increase the magnet current smoothly to three desired values between 0.9 and 1.5A (say 0.9, 1.2 & 1.5A). Keeping the glass block between the pole pieces of the magnet, the analyzer knob is slowly rotated to get an extinction of the transmitted light. Note down this reading. Now bring the power supply knob back to zero smoothly, switch off the power and reverse the direction of the magnet current. Switch on the power supply. Now smoothly increase the value of the current to the same value again. Rotate the analyzer knob to get extinction as before. Note this reading also. Let the two readings of analyzer for extinction be θ_i and θ_f respectively, then the Verdet constant V is given by

$$V = \frac{(\theta_f - \theta_i)}{nL\left(|B_f| \pm |B_i|\right)} = \frac{\Delta\theta}{2nLB_a} \quad , \tag{9}$$

where L is the dimension of the glass block along the axis of the magnet and n is the average refractive index of the glass for the light used. Here $|B_i|$ and $|B_f|$ denote the magnitude of the magnetic field values for the direct and reverse current (i.e. initial and final). B_a denotes the average value of magnetic field, i.e. $2B_a = |B_i| + |B_f|$.

NOTE: B_i and B_f may differ from each other due to hysteresis. Therefore current should be increased/decreased from/to zero slowly and smoothly in each adjustment to minimize the effect of hysteresis. The values of B_i and B_f would be known from the calibration curves for the magnet for direct and reverse currents obtained in section IIB above.

At least eight readings for $\Delta\theta$ should be taken by cycling the magnet current forward - backward - forward - backward in succession. (Devise a way to overcome backlash error in the analyzer). Record observations for three different fields (magnet current between 0.9 and 1.5A) and take average of $\Delta\theta$ / B_a and calculate V. There are two blocks of glass marked LF or EDF.

IV Questions

1. Could you suggest two other methods of measuring the magnetic field based on the experiments given in this manual?

2. How will θ change if you interchange your position with the source? Explain how this fact has been utilized in the above measurements.

3. What is the significance of the Faraday rotation experiment?

4. On what factors does the Verdet Constant of a solid, liquid, gas depend?

5. Show graphically the variation of the Verdet Constant with wavelength?

6. How would you distinguish the magneto-optical activity (i.e., as in Faraday rotation) from optical activity due to structure (such as in quartz, for example)?

Suggested reading

1. Read p. 101-106 of Optics by A Sommerfeld to understand the classical explanation of Faraday rotation.

2. Read p. 312-315 of Light Principles and Experiments by G.S. Monk to understand the derivation of the formula for θ (Eq. (2)) above.

V References

1. F.A.Jenkins and H.E.White, "Fundamentals of Optics".

2. S.G. Mavroidos, "Magneto-Optical Properties" in E.Abeles (Ed), "Optical Properties of Solids".

Useful Data :

$NA = (214 \pm 1)\ cm^2$

Length of the sample = L = (1.8±0.1) cm.

Refractive Index of Light Flint [LF] = 1.56±0.01 for sodium

Refractive Index of Extra Dense Flint [EDF] = 1.65±0.01 yellow light

Standard Results: $V_{LF} = 5.3 \times 10^{-4}$ degree/ gauss/cm

$V_{EDF} = 7.4 \times 10^{-4}$ degree/ gauss/cm